Proteomics, Metabolomics and Multi-Omics

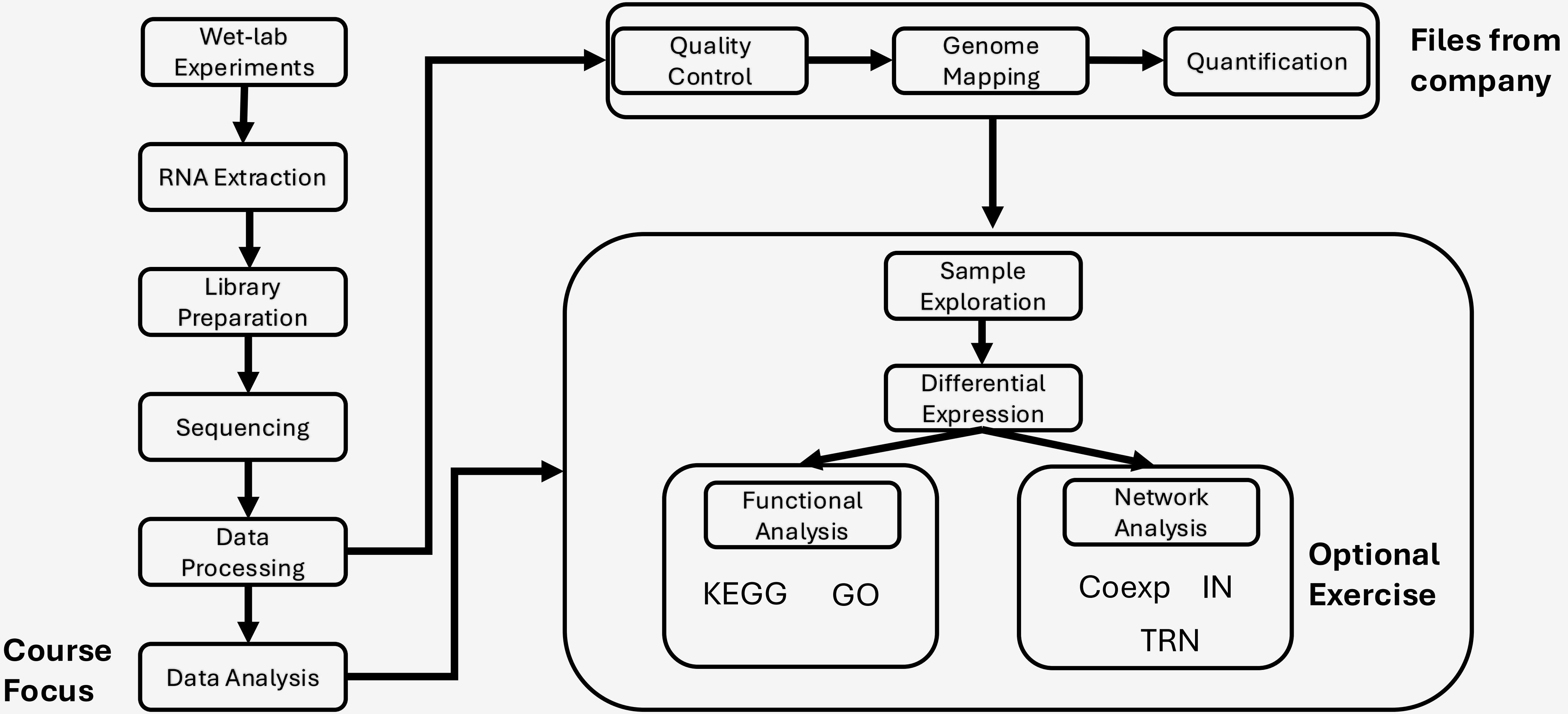
Muhammad Arif

muhammad.arif@gu.se

Computational Multi-Omics Lab

University of Gothenburg

Recap



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SciLifeLab

What becomes visible after RNA

Transcript → **Protein** → **Metabolite**

potential → machinery → biochemical state

Today is not “RNA-seq again”.
We focus on what proteins and metabolites reveal that RNA cannot.

What is new today?

PROTEOMICS

Protein abundance, turnover, complexes, isoforms, post-translational modifications, dynamic range.

METABOLOMICS

Small molecules, pathway output, diet/drug/environment signals, fast dynamics, annotation uncertainty.

INTEGRATION

When RNA, protein and metabolites agree, and when disagreement is the most interesting result.

We will NOT re-teach PCA, volcano plots and FDR.

Instead: what is biologically different, what can go wrong, and how to connect the layers.

Proteomics

Why RNA and protein disagree (and why that is useful)

mRNA → **translation** → **protein** → **function**

translation efficiency

protein half-life

localization / complex formation / PTMs

- A protein can change without its transcript changing.
- A transcript can change before the protein has time to respond.
- The same protein abundance can have very different activity states.

Discordance is not “bad data” by default - it can reveal regulation between omics layers.

Activity: RNA changed, what should the protein do?

Pairs, 2 minutes. Predict the protein-level outcome.

Gene A

RNA ↑ 8×
Protein half-life: 4 days
Sample collected 6 h after perturbation

Protein?

Gene B

RNA unchanged
Protein rapidly degraded after phosphorylation

Protein?

Gene C

RNA ↑ 2×
Translation strongly inhibited

Protein?

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mRNA → **translation** → **protein** → **function**

translation efficiency

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localization / complex formation / PTMs

Transcriptomics measures regulatory response. Proteomics measures the molecular machinery that survived translation + turnover

Proteomics: One "protein" is not one thing

ABUNDANCE

How much protein is present?

Most common entry point for differential proteomics.

ISOFORMS

Different protein forms can have different functions or localization.

PTMs

Phosphorylation, acetylation, ubiquitination etc. can change activity without changing abundance.

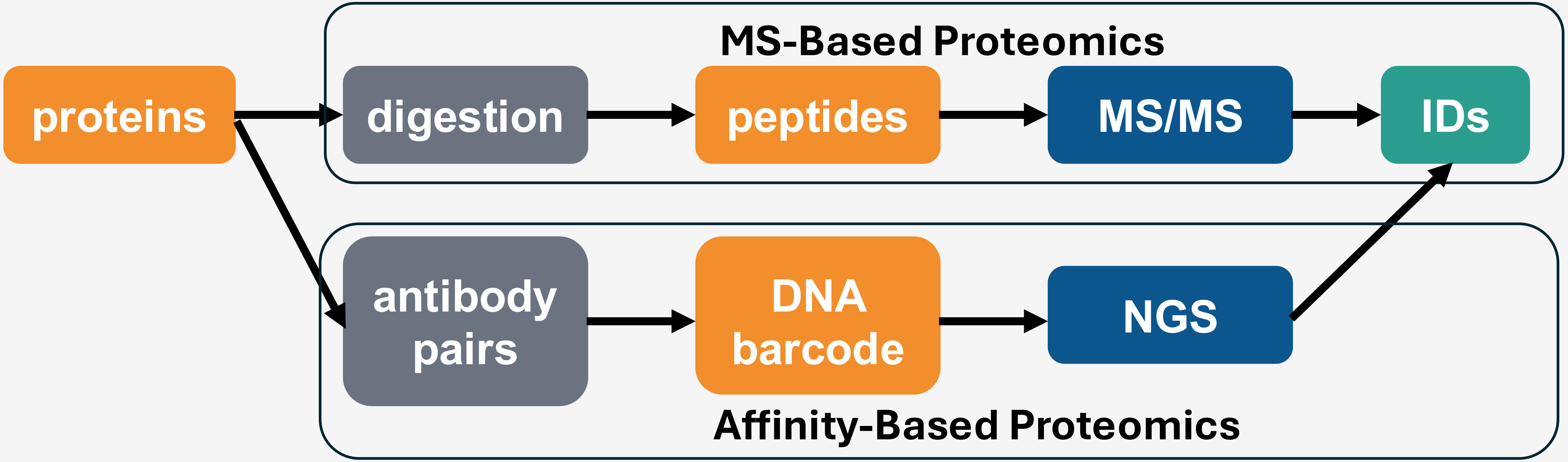
COMPLEXES

Function often depends on who the protein interacts with.

Example: kinase abundance unchanged
phosphorylation ↑ strongly
pathway activity ↑

Proteomics can move us closer to function than transcriptomics, but only if we know what "protein measurement" we actually have.

Measure or Infer Proteins?

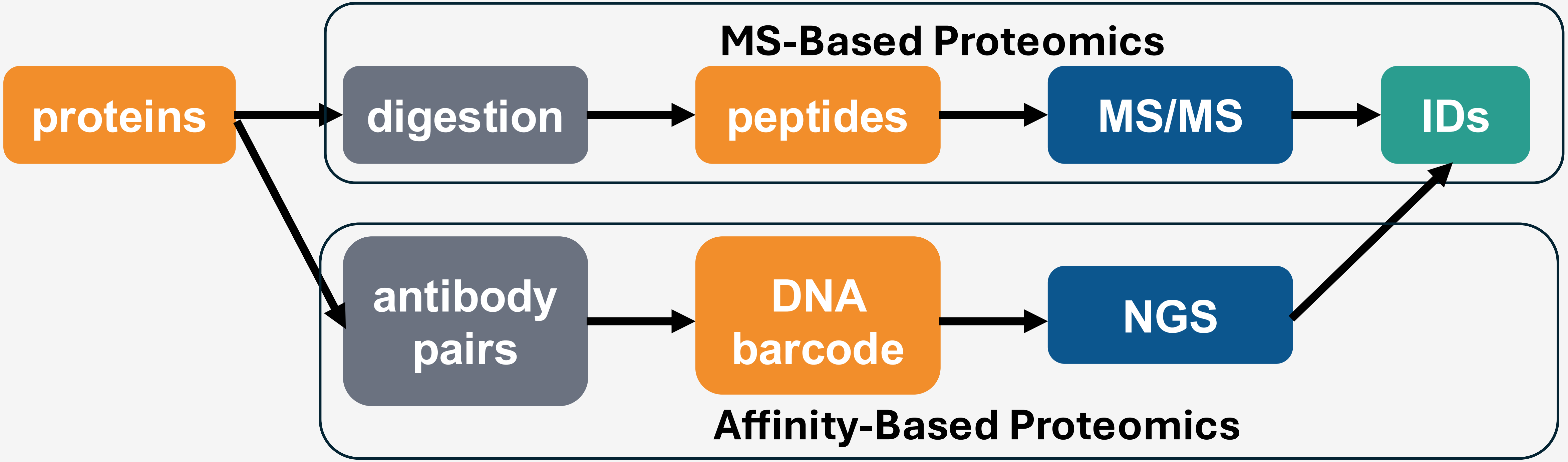


Protein inference
Several proteins can share peptides. Which protein produced the peptide?

Missing values
A missing intensity can mean "not detected", not necessarily "not present".

Dynamic range
Highly abundant proteins can dominate; low-abundance signaling proteins are harder.

Measure or Infer Proteins?



Proteomics has an identification problem before it has a statistics problem.

Proteomics Pipeline

Proteomics pipeline is similar with Transcriptomics!

load counts + metadata ▸ **check sample IDs** ▸ **filter low counts** ▸ **normalize / transform** ▸ **PCA** ▸ **DESeq2** ▸ **volcano + heatmap** ▸ **enrichment**

Protein intensities

Check Missingness

T-Test
Limma
MSstats
DEP
Mixed Models (LMER)

ENRICHMENT

GO / Reactome / KEGG

Similar to Transcriptomics

PROTEIN NETWORKS

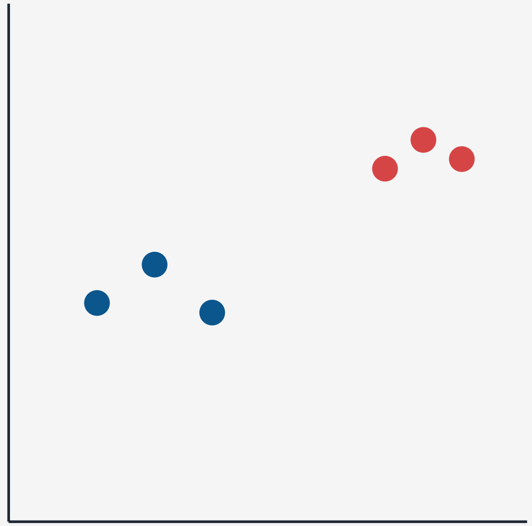
STRING / interaction networks
Do changed proteins form a connected module?

PTM ENRICHMENT

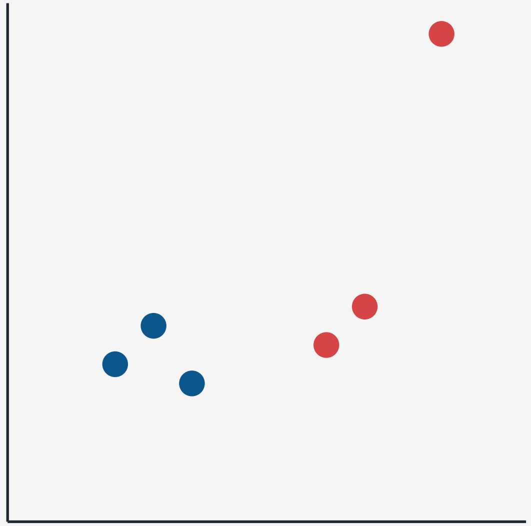
Kinase / phosphosite analysis
Which signaling programs may be active?

Proteomics QC: what would make you nervous?

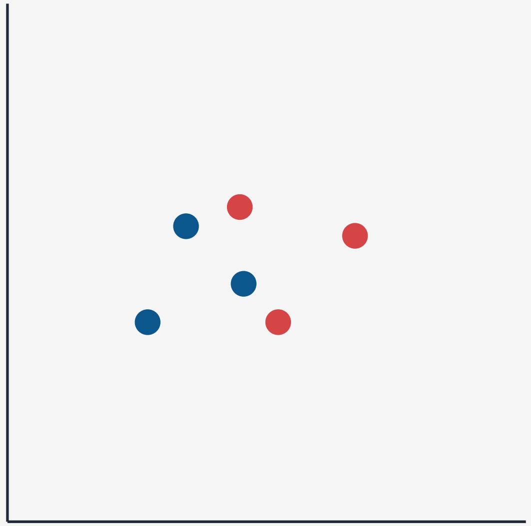
Quick vote: which plot would make you stop the analysis first?



A group separation



B one extreme sample

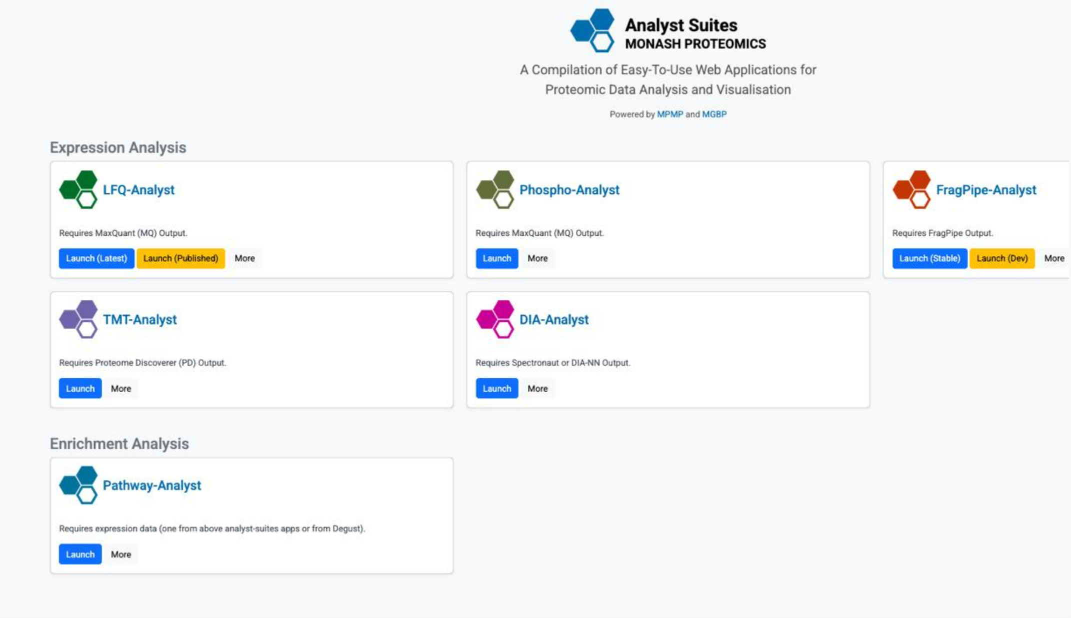


C no condition pattern

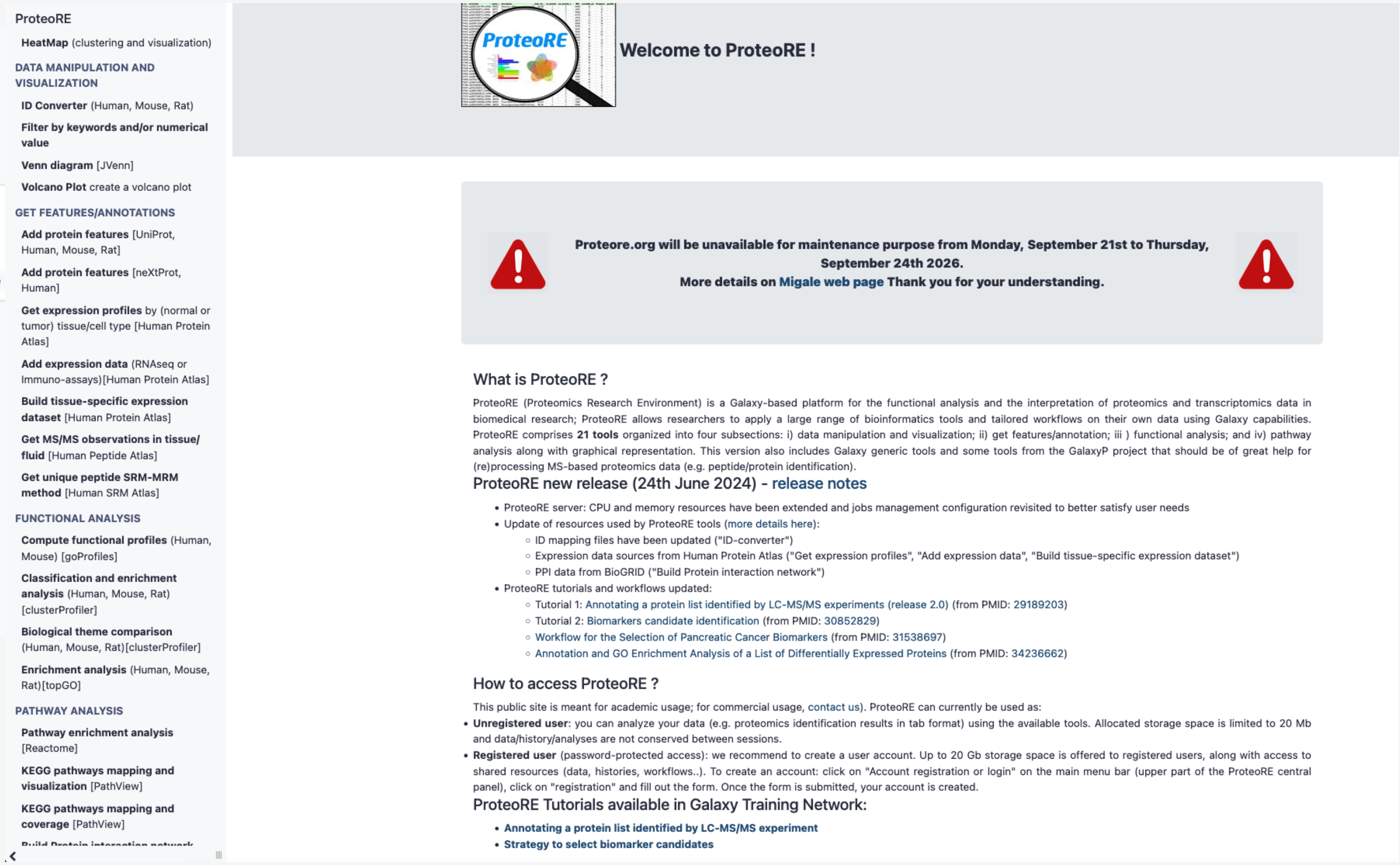
In proteomics, also inspect missingness per sample/protein and intensity distributions
PCA alone is not enough.

Online Tools

Analyst Suites
Monash Proteomics



ProteoRE



Metabolomics

Metabolomics: a biochemical snapshot

FAST
Metabolites can change in seconds or minutes.
They can respond much faster than transcription.

CONTEXTUAL
Diet, drugs, microbiome, fasting, circadian rhythm, handling and storage can all matter.

CLOSE TO PHENOTYPE
Metabolites are substrates/products of biochemical reactions and often reflect pathway state.

Think about Abundance vs Flux!

Glucose → **Pyruvate** → **Lactate**

steady-state abundance

metabolic flux

Concentration question
“Is lactate higher?”

Flux question
“Is more glucose carbon being converted into lactate?”

Targeted vs Untargeted

TARGETED

Measure a defined metabolite panel.

+ better identification confidence
+ often more quantitative
+ easier biological interpretation
− limited to what you decided to measure

UNTARGETED

Measure thousands of spectral features.

+ discovery potential
+ broad chemical space
− many features remain unknown
− one metabolite may create several peaks

Untargeted metabolomics often begins with “features” metabolite identity is a separate analytical step.

Questions for Metabolomics

EXPOSURE

Drug metabolites, diet-derived compounds, environmental chemicals.

ENERGY / REDOX

ATP/ADP, NAD-related metabolites, TCA intermediates, glutathione.

PATHWAY OUTPUT

Accumulation or depletion of substrates/products can reveal pathway bottlenecks.

A transcript can tell you an enzyme may be induced. A metabolite can tell you the biochemical system actually changed.

But remember: concentration ≠ flux. For flux, think isotope tracing.

Metabolomics Pipeline

Metabolomics pipeline is similar with Proteomics! Both starts from continuous abundance

load counts + metadata → **check sample IDs** → **filter low counts** → **normalize / transform** → **PCA** → **DESeq2** → **volcano + heatmap** → **enrichment**

Protein intensities

Metabolite intensities

Check Missingness

Imputation

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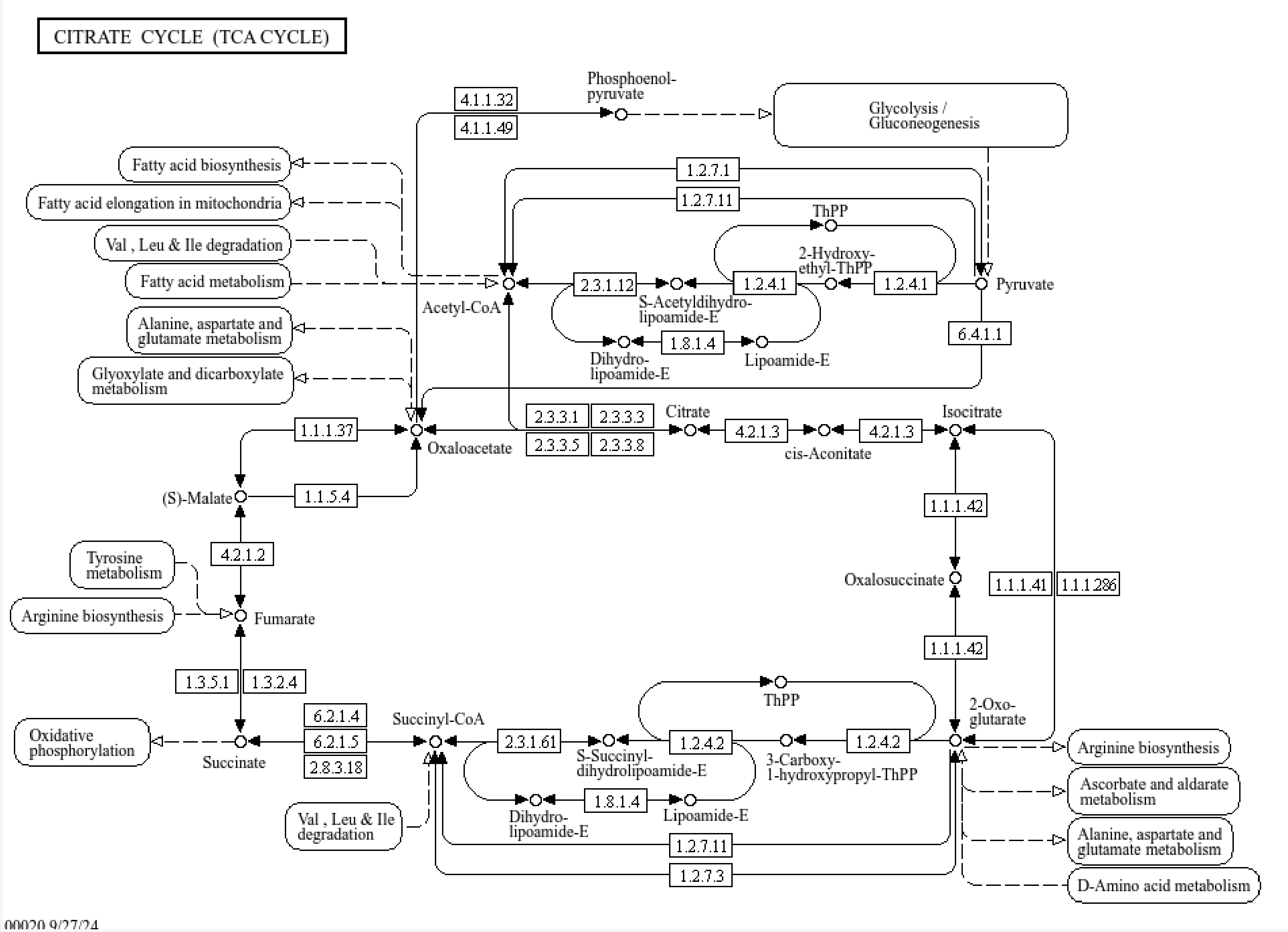
ENRICHMENT

~~GO~~ / Reactome / **KEGG/ MetaboAnalyst**

KEGG for Metabolomics



MetaboAnalyst

MetaboAnalyst 6.0 - from raw spectra to biomarkers, patterns, functions and systems biology

Module Overview

Input Data Type	**Available Modules** (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features (metabolite list or table)		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		

Things can get exciting: Omics Disagreement

Transcript	Protein	Metabolite	
↑	↑	↑	coherent activation
↑	↔	↑	translation delay / protein stable
↔	↑	↓	post-transcriptional regulation / substrate consumption
↓	↑	↑	compensation or protein stabilization

The most informative multi-omics result is often not agreement. It is a biologically interpretable mismatch.

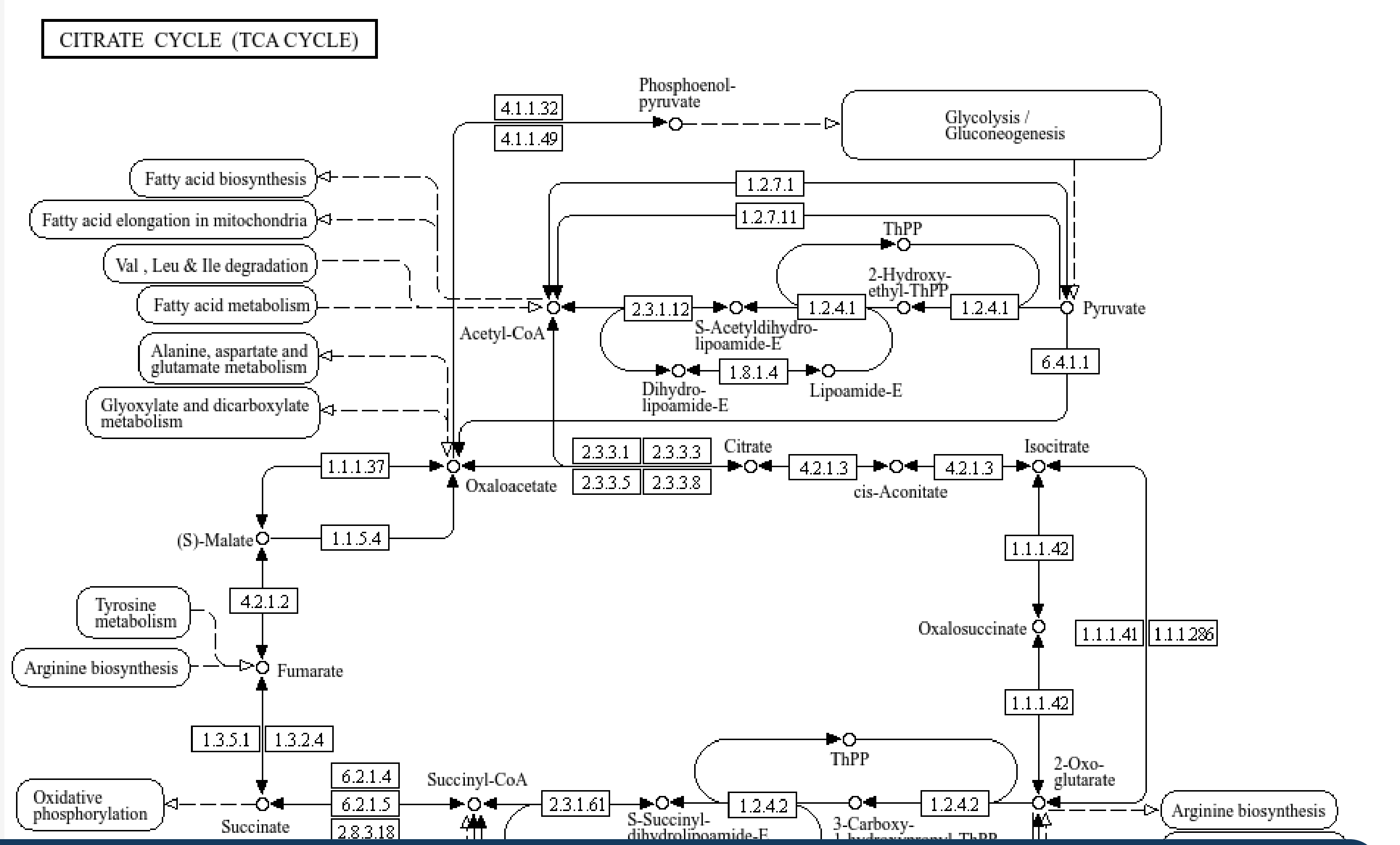
MultiOmics Integration

Simple ways to integrate Omics

1. PATHWAY CONVERGENCE

Analyze each omics separately. Ask whether changed genes/proteins/metabolites point to the same pathway.

Best for beginners.



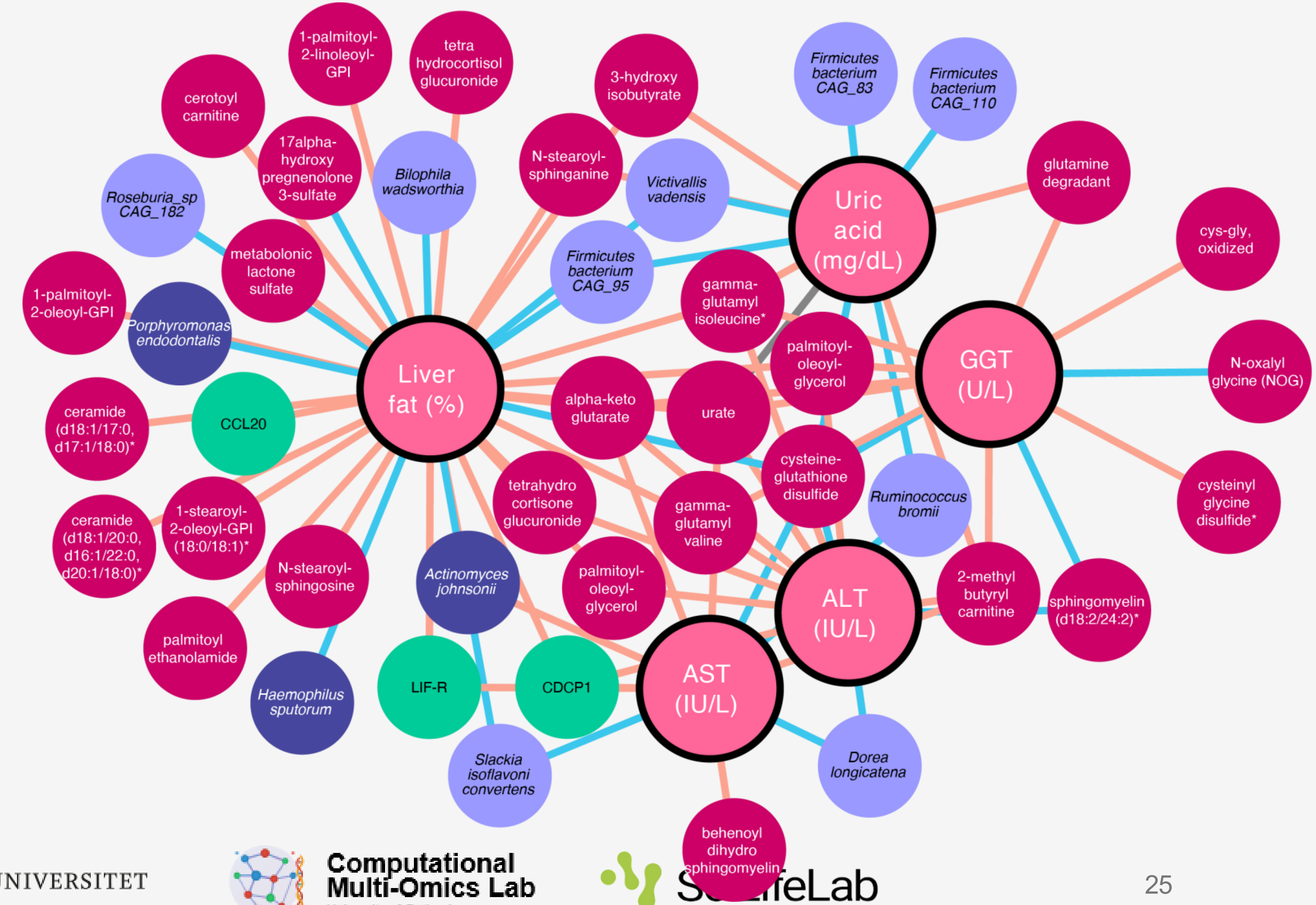
Star simple. Integration should answer a biological question, not just combine matrices because you have them.

Simple ways to integrate Omics

2. FEATURE CORRELATION

Match samples across layers. Correlate gene ↔ protein, enzyme ↔ metabolite, protein ↔ metabolite.

Good for hypothesis generation.



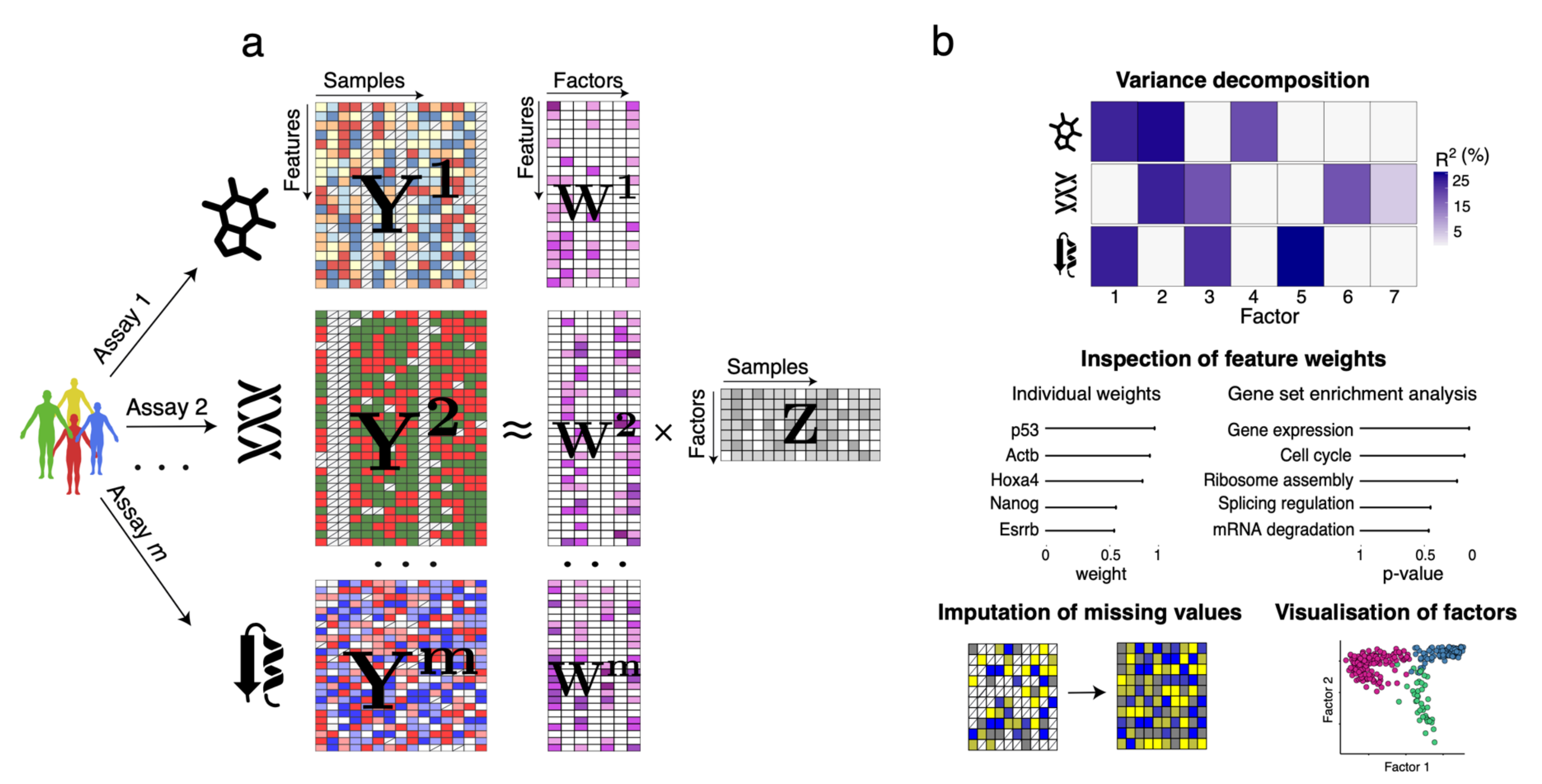
Simple ways to integrate Omics

3. JOINT MODELS

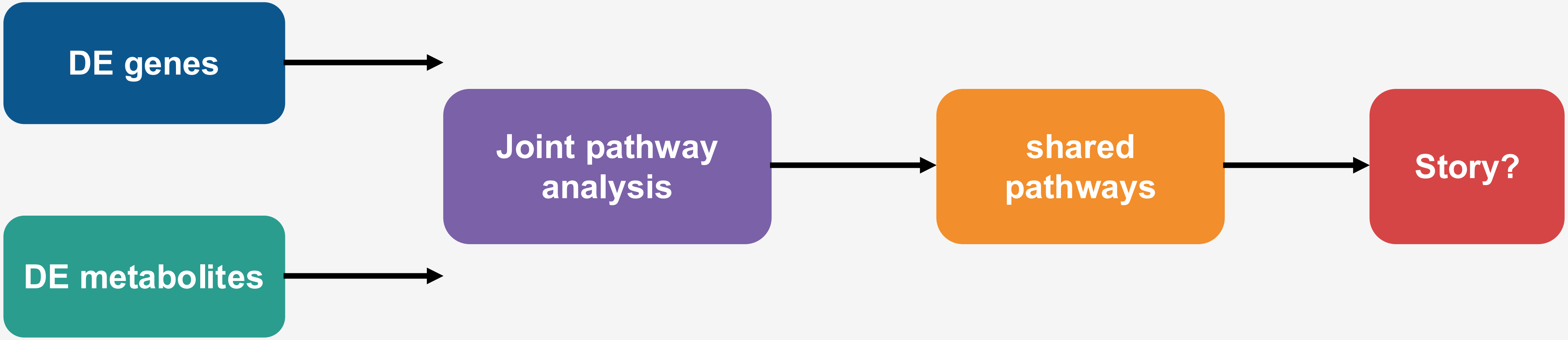
Integrate matrices together: latent factors, sparse models, networks, machine learning.

Powerful, but harder to interpret.

MOFA or MixOmics



Transcriptomics + Metabolomics



- Example: collagen/fibrosis genes ↑ + TCA/redox metabolites altered → stronger evidence for metabolic remodeling.
- But a “shared pathway” is not proof that the gene change caused the metabolite change.
- Use time, perturbation or targeted validation to move toward mechanism.

Integration strengthens interpretation when independent molecular layers converge on the same biology

Proteomics + Metabolomics

Enzyme protein → **Substrate** → **Product**

Pattern A

enzyme ↑, substrate ↓, product ↑
→ consistent with increased pathway throughput

Pattern B

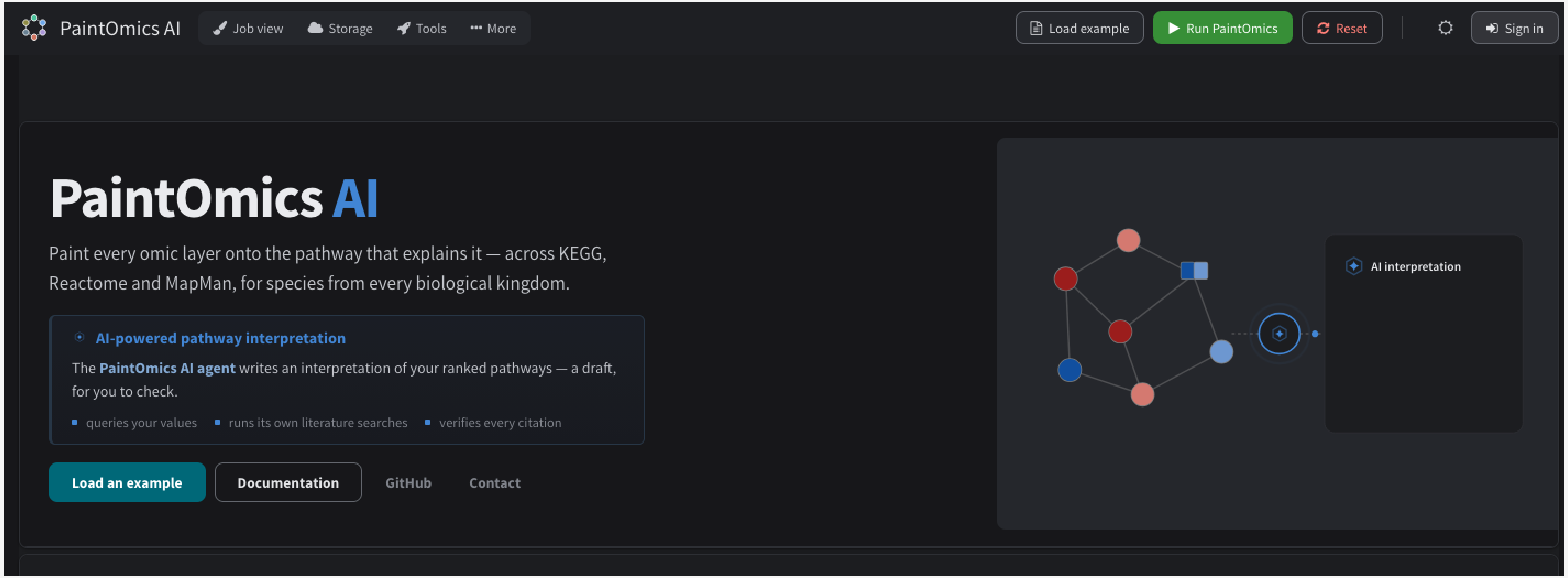
enzyme ↑, substrate ↑, product ↔
→ possible bottleneck / inactive enzyme / missing cofactor

Pattern C

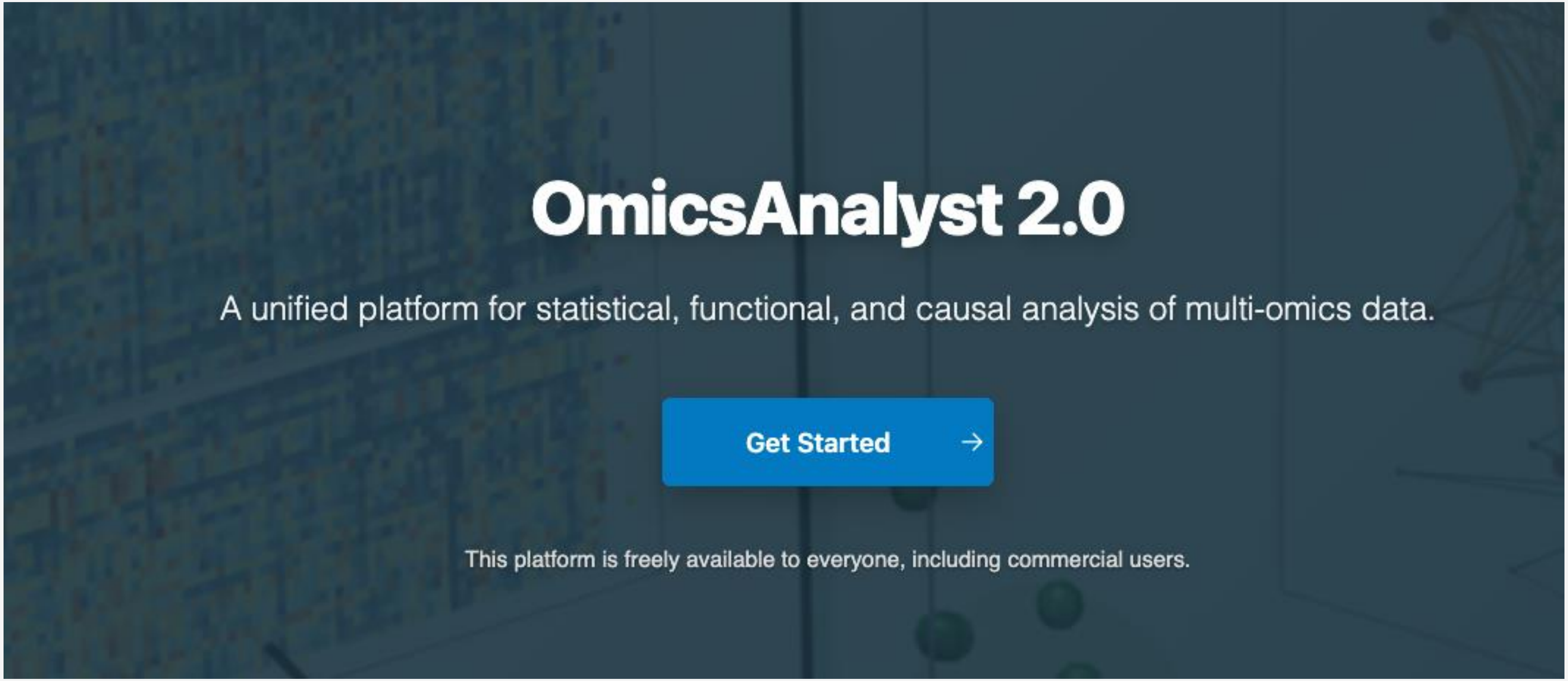
enzyme ↔, product ↑
→ upstream supply or regulation elsewhere

Multi-omics becomes mechanistic when you interpret molecules in reaction context, not only by correlation.

PaintOmics 4



OmicsAnalyst



So many tools and methods! How to choose?

Do Pathways Agree?

1. PATHWAY CONVERGENCE

Analyze each omics separately. Ask whether changed genes/proteins/metabolites point to the same pathway.

Best for beginners.

Which features co-vary?

2. FEATURE CORRELATION

Match samples across layers. Correlate gene ↔ protein, enzyme ↔ metabolite, protein ↔ metabolite.

Good for hypothesis generation.

I need to predict phenotypes

3. JOINT MODELS

Integrate matrices together: latent factors, sparse models, networks, machine learning.

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Take-home messages

PROTEOMICS

Closer to molecular machinery.
Think peptides, protein inference, missingness, PTMs and networks.

METABOLOMICS

Closer to biochemical phenotype.
Think annotation, scaling, exposure, pathway output and flux.

MULTI-OMICS

Agreement is useful.
Disagreement can be even more informative if biology explains it.

The statistical toolkit is transferable. The interpretation is omics-specific.

Ask: What does this layer measure that the previous layer could not?

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